

FORM – 2

THE PATENTS ACT, 1970

(39 of 1970)

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THE PATENTS RULES, 2003

COMPLETE SPECIFICATION

(See section 10 and rule 13)

**WEARABLE GLOVE SYSTEM FOR TRANSLATING SIGN LANGUAGE INTO
CONTEXT-AWARE AND EMOTIONALLY MODULATED SPEECH**

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**The following specification particularly describes the invention and the manner in which it
is to be performed.**

WEARABLE GLOVE SYSTEM FOR TRANSLATING SIGN LANGUAGE INTO CONTEXT-AWARE AND EMOTIONALLY MODULATED SPEECH

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FIELD OF THE DISCLOSURE

[0001] The present invention relates to a wearable system designed to translate sign language into speech. More specifically, it encompasses a glove-based system integrating flex sensors, an inertial measurement unit, a processing unit, and a communication module to capture, process, and convert hand and finger gestures
10 into contextually appropriate and emotionally expressive speech. The system supports selectable voice profiles and operational modes for professional and casual communication, aiming to enhance interaction and personal expression for speech-impaired users while ensuring accurate and natural speech synthesis.

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BACKGROUND

[0002] The subject matter discussed in the background section should not be assumed to be prior art merely as a result of its mention in the background section. Similarly, a problem mentioned in the background section or associated with the subject matter of the background section should not be assumed to have been
20 previously recognized in the prior art. The subject matter in the background section merely represents different approaches, which in and of themselves may also correspond to implementations of the claimed technology.

[0003] Current sign language translation technologies rely primarily on either software-only systems or basic glove-based hardware that detects finger positions
25 without contextual understanding. Such systems often generate flat, monotonic speech, lacking emotional expressiveness, personalization, and social adaptability. Users of these systems face communication barriers due to robotic outputs that fail to convey emotional tone or situational context. Additionally, conventional solutions do not offer selectable voice profiles or adaptive speech synthesis. The
30 need for a more sophisticated wearable system (100) that captures both static

handshapes and dynamic gestures, while producing speech modulated by emotional state and context, is crucial for enabling authentic, socially integrated communication for speech-impaired individuals.

5 **[0004]** Sign language involves both static postures and dynamic hand motions, including speed, acceleration, and intensity, which convey nuanced meaning and emotion. Conventional translation systems fail to detect or interpret these gesture dynamics accurately, resulting in incomplete information transfer. Users cannot express subtle emotional cues, limiting their ability to communicate effectively in both professional and casual environments. Furthermore, traditional devices often
10 lack personalization options such as selectable voice gender profiles, preventing users from establishing an identity-aligned communication style. Therefore, a system capable of capturing precise finger and hand movements through integrated flex sensors and IMU, while combining machine learning with contextual speech modulation, is necessary.

15 **[0005]** Existing wearable gloves do not provide adaptive translation according to situational requirements, nor do they incorporate advanced text-to-speech engines capable of prosody, pitch, and volume modulation based on gesture dynamics. Communication is limited to literal translations without considering professional versus casual contexts, leaving users with unnatural outputs. Additionally, most
20 systems rely on external computing for processing, reducing portability and real-time responsiveness. The invention overcomes these deficiencies by integrating a processing unit into the glove, coupled with a communication module to interface with mobile devices, thereby enabling on-the-fly interpretation, contextual mode selection, and emotional modulation of speech, ensuring authentic, expressive
25 communication for speech-impaired individuals.

[0006] According to a patent application “US20020152077A1” titled “Sign language translator” discloses A method and apparatus for translation of hand positions into symbols. A glove for wearing on an operator's hand includes bend sensors disposed along the operator's thumb and each finger. Additional bend
30 sensors are located between selected fingers and along the wrist. A processor generates a hand position signal using bend sensor signals read from the bend

sensors and transmits the hand position signal to an output device. The output device receives the hand position signal and generates a symbol representative of the hand position signal using the received hand position signal and a lookup table of hand position signals associated with a set of symbols. The output device then
5 produces either a visual or audio output using the symbol.

[0007] According to a patent application “US20160307469A1” titled “System and Method For Automated Sign Language Recognition” discloses A method for sign language recognition includes receiving, with an input device, an input based on a plurality of hand movements and postures of a user that correspond to a sequence
10 of signs, extracting a plurality of features from the input corresponding to the plurality of hand movements and postures, identifying, a start of the sequence of signs in the input based on a first set of features in the plurality of features and a first Hidden Markov Model (HMM) stored in the memory and identifying a first sign in the input based on a second set of features in the plurality of features and a
15 second HMM stored in the memory. The method also includes generating an output corresponding to the first sign from the input.

[0008] Therefore there is a need for provide a wearable system that captures precise finger and hand movements through flex sensors and an IMU for accurate sign language detection.

OBJECTIVES OF THE INVENTION

[0009] The objective of invention is to develop provide a wearable system that captures precise finger and hand movements through flex sensors and an IMU for accurate sign language detection.

5 [0010] The objective of invention is to generate context-aware speech output with selectable modes, including professional and casual/frank speech, enhancing social adaptability.

[0011] Furthermore, the objective of present invention is to integrate gesture-based emotional modulation for expressive speech, enabling users to convey urgency,
10 calmness, excitement, or other emotions naturally.

[0012] Furthermore, the objective of present invention is to provide personalized voice profiles (male, female, or gender-neutral) for user-specific speech synthesis.

SUMMARY

[0013] The invention is a wearable glove system that translates sign language into
15 speech with contextual and emotional awareness. Flex sensors capture static finger positions, while an inertial measurement unit measures dynamic hand motions. A processing unit digitizes and pre-processes the sensor data, applying machine learning for accurate gesture classification. A communication module transmits data to a paired smartphone or executes on-device synthesis. The system provides
20 selectable communication modes—professional and casual—and adjusts speech prosody, pitch, rate, and volume based on gesture-derived emotional intent. Voice personalization allows user-specific outputs, offering authentic, socially adaptable, and expressive communication for speech-impaired users.

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BRIEF DESCRIPTION OF THE DRAWINGS

[0014] The accompanying drawings illustrate various embodiments of systems, methods, and embodiments of various other aspects of the disclosure. Any person with ordinary skills in the art will appreciate that the illustrated element
5 boundaries (e.g., boxes, groups of boxes, or other shapes) in the figures represent one example of the boundaries. It may be that in some examples one element may be designed as multiple elements or that multiple elements may be designed as one element. In some examples, an element shown as an internal component of one element may be implemented as an external component in another, and vice versa.
10 Furthermore, elements may not be drawn to scale. Non-limiting and non-exhaustive descriptions are described with reference to the following drawings. The components in the figures are not necessarily to scale, emphasis instead being placed upon illustrating principles.

[0015] FIG. 1 illustrates a block diagram of a wearable sign language-to-speech
15 system, according to an embodiment of the present invention; and

[0016] FIG. 2 illustrates a pictorial representation of the wearable sign language-to-speech system, according to an embodiment of the present invention.

DETAILED DESCRIPTION

[0017] Some embodiments of this disclosure, illustrating all its features, will now be discussed in detail. The words “comprising,” “having,” “containing,” and “including,” and other forms thereof, are intended to be equivalent in meaning and
5 be open ended in that an item or items following any one of these words is not meant to be an exhaustive listing of such item or items, or meant to be limited to only the listed item or items. It must also be noted that as used herein and in the appended claims, the singular forms “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise.

10 [0018] Although any systems and methods similar or equivalent to those described herein can be used in the practice or testing of embodiments of the present disclosure, the preferred, systems and methods are now described. Embodiments of the present disclosure will be described more fully hereinafter with reference to the accompanying drawings in which like numerals represent like elements
15 throughout the several figures, and in which example embodiments are shown. Embodiments of the claims may, however, be embodied in many different forms and should not be construed as limited to the embodiments set forth herein. The examples set forth herein are non-limiting examples and are merely examples among other possible examples.

20 [0019] FIG. 1 illustrates a block diagram of a wearable sign language-to-speech system (100), according to an embodiment of the present invention. FIG. 2 illustrates a pictorial representation (200) of the wearable sign language-to-speech system, according to an embodiment of the present invention.

[0020] The wearable system (100) is a glove made from flexible, lightweight, and
25 non-conductive material. Flex sensors (102) are integrated along each finger and thumb to detect joint flexure. The IMU (104), comprising a 3-axis accelerometer and gyroscope, is mounted on the rear of the glove to capture linear and angular motion. A processing unit (106), housed in the wrist section, digitizes sensor data and performs preliminary pre-processing. A communication module (108)
30 transmits processed data to a paired device or receives commands. The glove includes a rechargeable lithium-ion battery supplying power to all components, enabling portable, real-time operation for sign-to-speech translation.

[0021] The flex sensors (102) continuously sample finger positions, while the IMU (104) records hand motion, orientation, and acceleration at high frequency. The processing unit (106) applies noise filtering, signal normalization, and data scaling to produce clean inputs for gesture recognition. Pre-processing ensures consistency across repeated gestures and accounts for variability in hand size, speed, and environmental factors. The communication module (108) forwards processed sensor signals to the paired device for further classification. This pipeline allows the system (100) to maintain real-time responsiveness, minimizing latency in detecting and interpreting sign language, ensuring accurate and timely translation into speech with minimal computational overhead.

[0022] The system (100) employs a machine learning model hosted on the paired device or on the processing unit (106) for gesture recognition. Pre-trained recurrent neural networks (RNNs) or convolutional neural networks (CNNs) classify incoming sensor data streams into corresponding signs. The model maps combinations of flex sensor (102) outputs and IMU (104) motion readings to vocabulary words, phrases, or phonemes. Continuous learning is supported by updating the model with new user-specific gestures. Accurate classification allows high fidelity translation of nuanced hand motions, laying the foundation for context-aware and emotionally expressive speech, a distinct improvement over prior mechanical or literal translation systems.

[0023] A software module on the paired device interprets recognized gestures and applies contextual processing. Users select a communication mode: professional or casual/frank. In professional mode, the NLP engine converts colloquial signs into formal equivalents suitable for business, interviews, or education. In casual mode, signs are translated directly into conversational expressions appropriate for informal communication. The processing unit (106) synchronizes mode selection with real-time sensor data, ensuring that each gesture triggers contextually relevant speech. This feature allows the system (100) to adapt translations to social situations, enhancing user confidence and social integration, while differentiating it from prior flat-output systems.

[0024] The system (100) derives emotional intent from gesture dynamics. Flex sensor (102) bending speed and IMU (104) motion intensity serve as proxies for

emotions such as excitement, calmness, or urgency. The processing unit (106) transmits this data to a TTS engine, which adjusts prosody, pitch, rate, and volume in real-time. High acceleration and rapid movements correlate with heightened emotion, while slower, smooth gestures indicate subdued emotions. By integrating emotional modulation, the system (100) provides expressive speech, enabling users to convey nuanced feelings. This capability significantly improves authenticity and enriches communication compared to robotic prior art solutions.

[0025] The system (100) supports selectable voice profiles, including male, female, and gender-neutral voices. Users configure the voice in the paired device interface, and the processing unit (106) ensures that all synthesized speech adheres to the chosen profile. Combined with emotional modulation, this personalization enables tailored communication experiences. Voice profiles are stored securely in the paired device and retrieved during real-time speech synthesis. The system (100) thereby provides users with identity-aligned outputs, enhancing comfort, agency, and social integration. This feature distinguishes the invention from conventional gloves lacking configurable voice characteristics.

[0026] The processing unit (106) processes data streams from flex sensors (102) and IMU (104) locally or offloads computations to the paired device. The communication module (108) employs Bluetooth Low Energy (BLE) for low-latency transmission. Sensor fusion and pre-processing reduce bandwidth requirements, allowing continuous real-time operation. Data security and low power consumption are maintained, enabling extended use without frequent recharging. The integration of local and paired-device processing ensures that the system (100) maintains rapid response times, supporting natural conversation and immediate speech output for dynamic hand gestures.

[0027] Processed gestures are converted into speech through a TTS engine on the paired device or embedded in the glove (100). The engine incorporates contextual mode selection, emotional modulation, and voice profile settings to produce authentic, expressive speech. Output may be transmitted via the paired device's speaker or a miniature speaker in the wrist enclosure. Speech timing aligns with hand movements, ensuring coherence between gestures and auditory output. This real-time feedback improves user experience and enables socially appropriate

communication, overcoming limitations of conventional devices that provide delayed or robotic responses.

[0028] The system (100) is suited for speech-impaired users in professional, educational, and casual settings. By converting gestures into context-aware, emotionally expressive speech, users achieve authentic self-expression and social integration. Voice personalization enhances identity representation, while mode selection ensures situational appropriateness. Real-time translation and emotional modulation improve communication fluidity, reducing social friction. The system (100) supports learning and adaptation, allowing users to expand vocabulary and gestures over time. Compared to existing mechanical or software-only solutions, this glove facilitates richer, more nuanced communication and provides an inclusive tool for professional and casual interactions.

[0029] The system (100) uniquely combines flex sensors (102), IMU (104), a processing unit (106), and communication module (108) with machine learning, contextual mode selection, emotional modulation, and voice personalization. It produces socially appropriate, expressive speech in real-time, overcoming robotic and literal translations of prior art. Users gain enhanced self-expression, identity representation, and professional integration. The glove's portability, low power consumption, and real-time operation ensure practicality and convenience. Emotional and contextual awareness enables richer communication, while high gesture recognition accuracy ensures reliability. This integrated approach surpasses existing solutions in authenticity, adaptability, personalization, and user empowerment.

[0030] It may have been described in detail through specific components, embodiments, and preferred implementations; however, it should be understood that the present invention is not limited to these particular examples. Modifications, variations, and equivalents can be made by those skilled in the art without departing from the scope or spirit of the invention as defined by the appended claims. The described system (100) provides a practical, context-aware, and emotionally expressive solution for translating sign language into speech, enhancing communication, social integration, and self-expression for speech-impaired individuals across professional and casual environments.

Dated Thisday of October, 2025

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CLAIMS

We Claim:

1. A wearable sign language-to-speech system (100) comprising:
 - a glove worn on one or both hands of a user,
 - 5 wherein the system (100) includes flex sensors (102) embedded along the length of each finger and thumb to detect finger joint flexure.
2. The system (100) of claim 1, further comprising an inertial measurement unit (IMU) (104) positioned on the glove, including a 3-axis accelerometer and a 3-axis gyroscope, for detecting linear and angular motion of the hand.
- 10 3. The system (100) of claim 2, further comprising a processing unit (106) in communication with the flex sensors (102) and IMU (104), wherein the processing unit (106) digitizes, pre-processes, and sums up the sensor data.
4. The system (100) of claim 3, further comprising a communication module (108) coupled to the processing unit (106) for transmitting pre-processed sensor
15 data to a paired computing device or receiving commands from it.
5. The system (100) of claim 3, wherein the processing unit (106) applies machine learning algorithms to classify the pre-processed data into corresponding sign language words, phrases, or phonemes.
6. The system (100) of claim 5, wherein a software module on the paired
20 computing device interprets the classified data and performs contextual processing for selecting at least one communication mode, including professional mode and casual/frank mode.
7. The system (100) of claim 6, further comprising an emotional modulation module configured to determine the user's emotional state from gesture dynamics,

including velocity and force, and modify the speech output prosody, pitch, rate, and volume accordingly.

8. The system (100) of claim 6, wherein the speech output is synthesized using a selectable voice profile, including male, female, or gender-neutral voices, to
5 provide personalized communication.

9. The system (100) of claim 1, further comprising a rechargeable power source configured to provide energy to the flex sensors (102), IMU (104), processing unit (106), and communication module (108), wherein the power source is positioned in the wrist section of the glove.

10 10. The system (100) of claim 1, wherein the combination of flex sensors (102), IMU (104), processing unit (106), communication module (108), contextual mode selection, emotional modulation, and voice profile selection enables authentic, context-aware, and emotionally expressive speech generation from sign language gestures, enhancing communication for speech-impaired users.

15 **Dated Thisday of October, 2025**

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ABSTRACT

WEARABLE GLOVE SYSTEM FOR TRANSLATING SIGN LANGUAGE INTO CONTEXT-AWARE AND EMOTIONALLY MODULATED SPEECH

5 A wearable glove system (100) translates sign language into contextually appropriate and emotionally expressive speech. Flex sensors (102) detect finger joint flexure, while an inertial measurement unit (104) captures hand motion dynamics. A processing unit (106) digitizes, pre-processes, and applies machine learning algorithms to classify gestures. A communication module (108) transmits
10 data to a paired device or embedded TTS engine. The system supports selectable modes, including professional and casual, and adjusts speech prosody, pitch, rate, and volume based on gesture-derived emotional intent. Users can select personalized voice profiles (male, female, gender-neutral) for tailored output. Real-time processing ensures responsive and accurate speech generation. The
15 system enables authentic communication, enriched with social context and emotional expression, enhancing self-expression, social integration, and user empowerment for speech-impaired individuals.

<FIG.1>

Dated Thisday of October, 2025

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